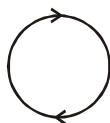


ELECTROSTATIC POTENTIAL & CAPACITANCE

1. A non-conducting solid sphere of radius R is uniformly charged. The magnitude of electric field at a distance r away from the centre.

- (1) Increases as r increases, for $r < R$
- (2) Decreases as r increases, for $0 < r < \infty$
- (3) Decreases as r decreases, for $R < r < \infty$
- (4) Is discontinuous at $r = R$

2. Given diagram can never represent



- (1) An electrostatic field
 - (2) A magnetostatic field
 - (3) Field for moving charge
 - (4) An induced electric field
3. Which of the following quantities depend on the choice of zero potential or zero potential energy.

- (1) Potential at a point
- (2) Potential difference between two points
- (3) Change in potential energy
- (4) Potential energy of a system of two charges

4. Under the influence of the Coulomb field of charge $+q$, a charge $-q$ is moving around it on an elliptical orbit. Find out the correct statements -

- (1) The angular momentum of the charge $-q$ is constant
- (2) The linear momentum of the charge $-q$ is constant
- (3) The angular velocity of the charge $-q$ is constant
- (4) The linear speed of the charge $-q$ is constant

5. Two equal charges q are placed at a distance of $2a$ and a third charge $-2q$ is placed at the midpoint. The potential energy of the system is

- (1) $\frac{q^2}{8\pi\epsilon_0 a}$
- (2) $\frac{6q^2}{8\pi\epsilon_0 a}$
- (3) $\frac{-7q^2}{8\pi\epsilon_0 a}$
- (4) $\frac{9q^2}{8\pi\epsilon_0 a}$

6. A conductor with a positive charge -

- (1) Is always at positive potential
- (2) Is always at zero potential
- (3) Is always at negative potential
- (4) May be +ve, zero or -ve potential

7. If a unit positive charge is taken from one point to another point over an equipotential surface then -

- (1) Work is done on the charge by electric field
- (2) Work is done by the charge against the electric field
- (3) Work done is non-zero constant by the field
- (4) No work is done by the field

8. The electric field inside a uniformly charged spherical shell is

- (1) Zero
- (2) Constant, less than zero
- (3) Directly proportional to the distance from centre
- (4) None of the above

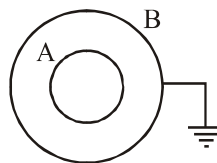
9. Two charged spheres of radii 10 cm and 15 cm are connected by a thin wire. No current will flow, if they have -

- (1) The same charge on each
- (2) The same potential
- (3) The same energy
- (4) The same field on their surfaces

10. Two small spheres each carrying a charge q are placed r metre apart. If one of the spheres is taken around the other one in a circular path of radius r , the work done by electric field will be equal to

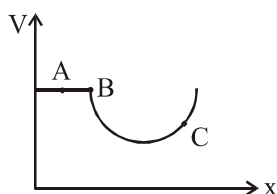
- (1) Force between them $\times r$
- (2) Force between them $\times 2\pi r$
- (3) Force between them $/ 2\pi r$
- (4) zero

11. A and B are two concentric metallic shells. If A is positively charged and B is earthed, then electric -



- (1) Field at common centre is non-zero
- (2) Field outside B is non-zero
- (3) Potential outside B is positive
- (4) Potential at common centre is positive

12. A metallic sphere is given some charge. Electric energy is stored.
- Only inside the sphere
 - Only outside the sphere
 - Both inside and outside
 - Neither inside nor outside
13. A hollow charged metallic spherical shell has radius R . If the potential difference between its surface and a point at a distance $3R$ from the centre is V then the value of electric field intensity at a point at distance $4R$ from the centre is -
- $\frac{3V}{19R}$
 - $\frac{V}{6R}$
 - $\frac{3V}{32R}$
 - $\frac{3V}{16R}$
14. Which is the correct statement?
- If the electric field due to a point charge varies as $r^{-2.5}$ instead of r^{-2} , then the Gauss law will still be valid
 - The Gauss law can be used to calculate the field distribution around an electric dipole
 - If the electric field between two point charges is zero somewhere, then the signs of the two charges are same
 - Work done by external force in moving a charge is always $(V_B - V_A)$
15. An imaginary graph is plotted between potential and distance then ($\vec{E}$ = electric field)



- At point A, $E = +ve$
 - At point C, $E = -ve$
 - At point C, $E = +ve$
 - At point C, $E = 0$
16. Work done against electrostatic force in bringing a unit positive charge from infinity to a point is:
- Equal to the electrostatic potential (V) at that point
 - equal to the negative of work done by electrostatic forces
 - Both (1) & (2)
 - Neither (1) nor (2)

17. If δW is the work done against electrostatic force in bringing of a infinitesimal small test charge δq from infinity to a point P, then potential at point P is

$$(1) \frac{\delta W}{\delta q} \quad (2) -\frac{\delta W}{\delta q} \quad (3) \delta W \cdot \delta q \quad (4) \text{None}$$

18. Work done in moving a charge from one point to the other inside a uniformly charged conducting sphere is -

- Always zero
- Non zero
- May be zero
- None of these

19. For linear isotropic dielectric the polarisation is

$$(1) p = \chi_e E \quad (2) p = -\chi_e E$$

$$(3) p = 2\chi_e E \quad (4) p = \frac{\chi_e}{E}$$

20. The total dipole moment of dielectric with polar molecules in absence of an external electric field is

- Zero
- Negative
- Infinite
- None of these

21. The extent of polarisation depends on -

- The dipole potential energy in the external field tending to align the dipole with the field
- Thermal energy tending to disrupt the alignment
- Both (1) and (2)
- Neither (1) nor (2)

22. The maximum electric field that a dielectric medium of a capacitor can withstand without break down is called its

- Polarisation
- Capacitance
- Dielectric strength
- None of these

23. If dielectric constant and dielectric strength be denoted by k & x respectively, then a material suitable for use as a dielectric in a capacitor must have

- high k & high x
- high k & low x
- low k & high x
- low k & low x

24. A parallel plate capacitor has a uniform electric field E ($V m^{-1}$) in the space between the plates. If the distance between the plates is d (m) and area of each plate is A (m^2) the energy (Joule) stored in the capacitor, is :

$$(1) \frac{1}{2} \epsilon_0 E^2 \quad (2) \epsilon_0 E A d$$

$$(3) \frac{1}{2} \epsilon_0 E^2 A d \quad (4) E^2 A d / \epsilon_0$$

25. A small sphere of radius r_1 having charge q_1 is enclosed by a spherical shell of radius r_2 having charge q_2 . Which charge will necessarily flow from the sphere to the shell, when connected.

(1) q_1 (2) q_2
(3) both (1) and (2) (4) May be q_2

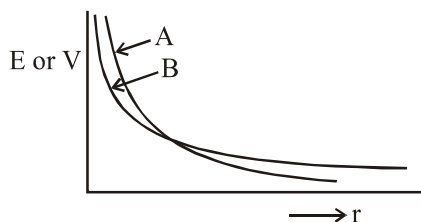
26. Van de graaff generator is used to

(1) Store electrical energy
(2) Build up high voltage of few million volts
(3) decelerate charged particle like electrons
(4) Both (1) & (2) are correct

27. The plates of a parallel plate capacitor are not exactly parallel. The surface charge density

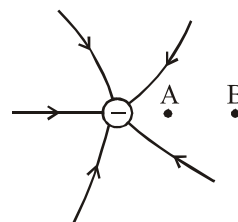
(1) Is lower at the closer end
(2) Will not uniform
(3) Each plate will have the same potential at every point
(4) Both (2) & (3)

28. Curve A and B represents variation in electric field and potential of a point charge then



- (1) A represents electric field
(2) B represents electric field
(3) Any of the curve may represent electric field and potential
(4) Can't be determined
29. Two charge $6 \mu\text{C}$ and $-4 \mu\text{C}$ are separated at distance 15 cm apart then the point on the line joining the charges where electric potential is zero is :
- (1) At distance 6 cm from $6 \mu\text{C}$
(2) At distance 45 cm from $6 \mu\text{C}$
(3) At distance 30 cm from $6 \mu\text{C}$
(4) At distance 60 cm from $6 \mu\text{C}$

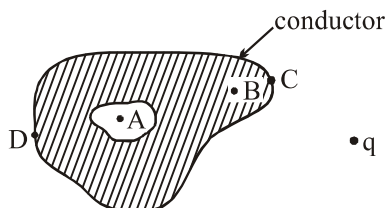
30. Figure shows the field lines of negative point charge then



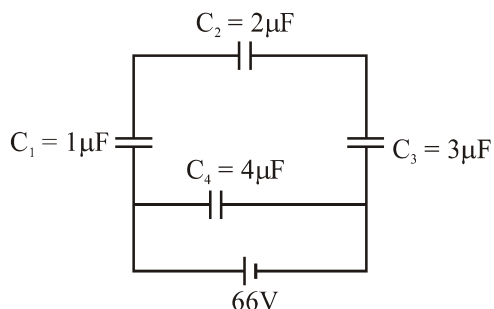
- (1) Sign of $V_B - V_A$ is negative
(2) Sign of work done by external agency in moving a negative charge from B to A is negative
(3) Sign of work done by the field in moving a positive charge from A to B is positive
(4) Sign of work done by the external agency in moving a negative charge from B to A is positive
31. A molecule of a substance has dipole moment 10^{-29} C-m. One mole of this substance is polarised by applying a strong electric field 10^5 V/m. If direction of electric field is suddenly changed by an angle 60° then heat released by the substance for the new alignment of the dipoles.
- (1) 0.3 J (2) 0.6 J
(3) 3 J (4) 6 J
32. Which of the following is not correct regarding a conductor
- (1) Inside a conductor, electrostatic field is zero
(2) Electrostatic field must be normal to the charged conducting surface
(3) Electric potential is necessarily constant throughout the volume of the conductor
(4) Electrostatic field at a point on the surface of

the charged conductor is $\frac{\sigma}{\epsilon_0}$

33. A cavity is cut inside a conductor and a charge q is placed outside it as shown in figure then incorrect statement is



- (1) Electric field at point A due to induced charges is zero
 (2) Electric field at point B due to the point charge q is non zero
 (3) σ_c is negative
 (4) Electric potentials at A and B are same
34. An air capacitor has capacitance $70 \mu\text{F}$ with a separation between plates 20 mm . Now distance between plates is doubled and dielectric of $k = 2$ thickness 10 mm is inserted between the plates of capacitor then its capacitance becomes.
 (1) $35 \mu\text{F}$ (2) $40 \mu\text{F}$
 (3) $50 \mu\text{F}$ (4) Remains unchanged
35. In the given figure charge on capacitor C_2 is



- (1) $22 \mu\text{C}$ (2) $44 \mu\text{C}$
 (3) $18 \mu\text{C}$ (4) $36 \mu\text{C}$

36. A metal sphere of radius R is given a potential V then energy stored inside the concentric sphere of radius $2R$ is

- (1) $\pi\epsilon_0 R V^2$ (2) $\frac{1}{2}\pi\epsilon_0 R V^2$
 (3) $2\pi\epsilon_0 R V^2$ (4) $4\pi\epsilon_0 R V^2$

37. For equipotential surfaces incorrect statement is

- (1) Are closer in region of stronger electric field
 (2) never be equally spaced
 (3) Perpendicular to electric field lines
 (4) Are more crowded near sharp edges of a conductor

38. A cylindrical capacitor has length 54 cm and radii 4 cm and 3 cm . Potential of the inner cylinder when it is given a charge 1 nC and outer cylinder is grounded

- (1) 10^4 volt (2) 10^3 volt
 (3) 10^5 volt (4) 10^6 volt

39. Two metal spheres of radii R and $2R$ having some charge density σ , are brought in contact then charge densities becomes

- (1) $\frac{\sigma}{3}, \frac{2\sigma}{3}$ (2) $\frac{2\sigma}{3}, \frac{\sigma}{3}$
 (3) $\frac{5\sigma}{3}, \frac{5\sigma}{6}$ (4) $\frac{5\sigma}{6}, \frac{5\sigma}{3}$

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	1	1	3	4	4	1	2	4	4	2	3	3	2	1	1	1	1	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	
Ans.	3	3	1	3	1	2	4	1	2	4	1	3	1	2	4	1	2	1	3	